Lecture 10

EE3010: Electrical Devices and Machines

School of Electrical and Electronic Engineering

Associate Professor Lee Peng Hin
Tel: +65 6790 4474 | Email: ephlee@ntu.edu.sg

Learning Objectives

By the end of this lecture, you should be able to:

- Describe the equivalent circuit of a DC generator.
- Derive and explain the voltage-current characteristics of a separately excited DC generator.
- Explain the methods used to control the terminal voltage of a separately excited DC generator.
- Analyse the separately excited DC generator, using the magnetisation curve and the equivalent circuit.

Introduction to DC Generators

- DC generators are DC machines used as generators.
- There is no real difference between a DC generator or motor, except for the direction of power flow.
- The various types of DC generators available differ in their terminal (voltage-current) characteristics, and therefore, in the applications to which they are suited.

Introduction to DC Generators

- In this course, the following types DC generators will be discussed in detail:
  - Separately excited DC generator – the field flux is derived from a separate power source and independent of the generator itself.
  - Shunt generator – the field flux is derived by connecting the field circuit directly across the terminals of the generator.

Introduction to DC Generators

- DC generators are compared by their voltages, power ratings, efficiencies, and voltage regulations defined as:

$$\text{VR} = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\%$$

$$V_{nl} = \text{no-load terminal voltage}$$

$$V_{fl} = \text{full-load terminal voltage}$$

- VR is the percentage change in terminal voltage, when the load is removed from the generator.

- DC generators are driven by a source of mechanical power called the prime mover of the generator. It can be a steam engine, diesel engine, or motor.

Example 1

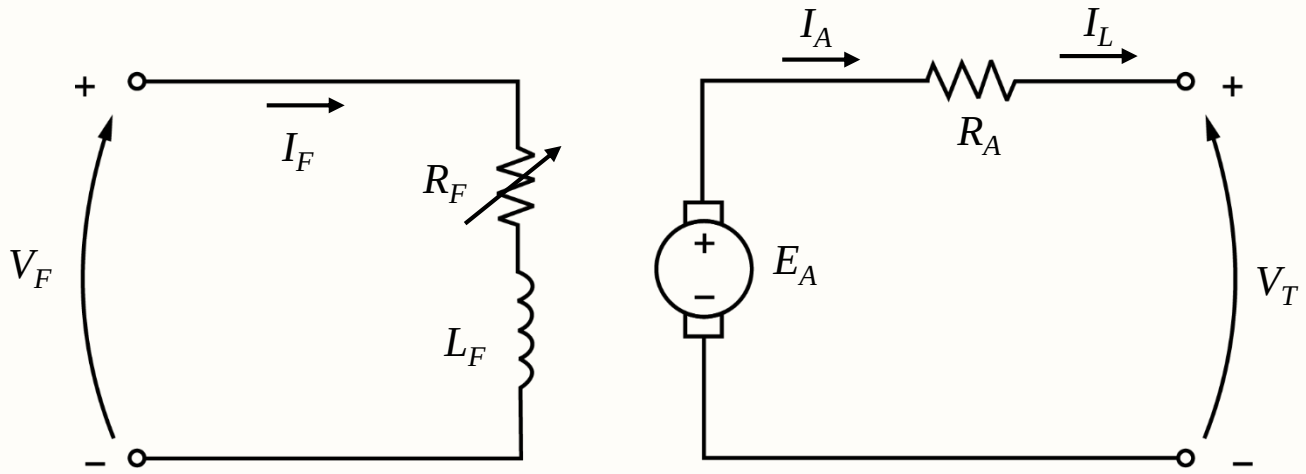
A 100 kW, 1800 r/min DC generator operating at rated load has a terminal voltage of 240 V. If the voltage regulation is 2.3 percent, determine the no-load voltage.

$$\mathrm{VR} = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\%$$

$$\Rightarrow 0.023 = \frac{V_{nl} - 240}{240}$$

$$\Rightarrow V_{nl} = 245.52 \text{ V}$$

Separately Excited DC Generator



Equivalent circuit of a separately excited generator.

R_F - combination of the resistance of the field windings and the external variable resistor, if any.

I_L - the current flowing in the lines connected to the terminals of the generator.

$$I_A = I_L,\ V_T = E_A - I_A R_A,\ I_F = \frac{V_F}{R_F}$$

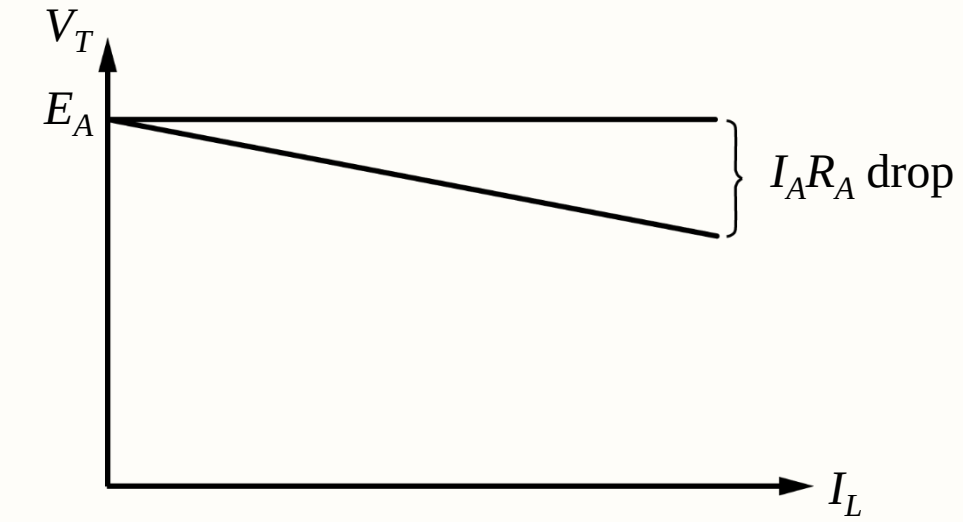
Example 2

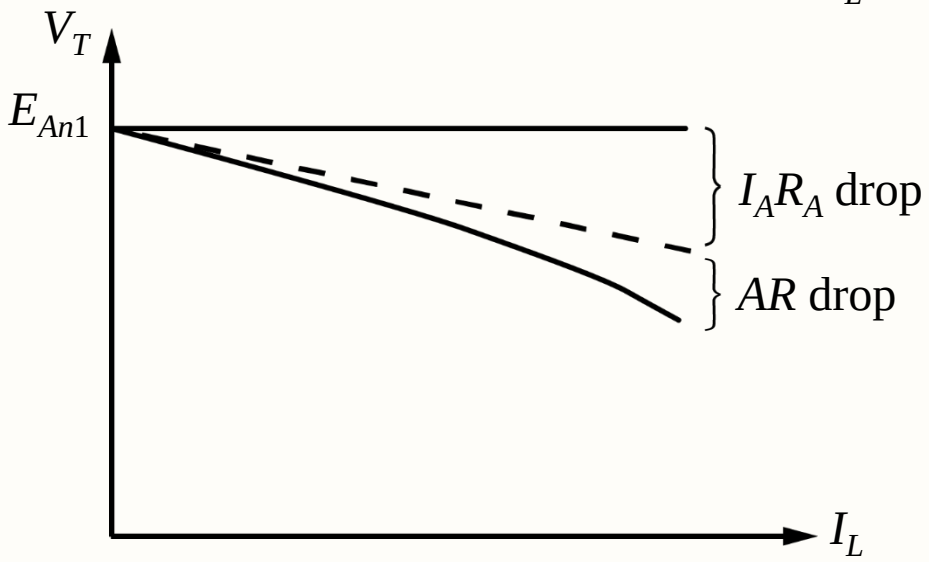
Determine the internal generated voltage, at rated conditions, of a separately excited generator which is rated at 25 kW, 250 V and 1450 r/min and has the following parameters.

$$R_A = 0.1053\Omega,\ R_{\text{interpoles}} = 0.0306\Omega,$$
$$R_{\text{comp windings}} = 0.0141\Omega,\ R_F = 96.3\Omega$$

$$I_A = \frac{P}{V_T} = \frac{25000}{250} = 100\ \text{A}$$
$$E_A = 250 + 100\left(0.1053\ + 0.0306 + 0.0141\right) = 265\ \text{V}$$

V-I Characteristics of a Separately Excited DC Generator





- When load supplied by generator ↑

  $\Rightarrow I_L$ (hence I_A) ↑

  $\Rightarrow I_A R_A \uparrow \Rightarrow V_T \downarrow$

- Armature reaction present (no compensating windings)

  $\Rightarrow$ flux weakening

  $\Rightarrow E_A = K\left(\phi\downarrow\right)\omega_m \downarrow$

  $\Rightarrow V_T \downarrow$ further

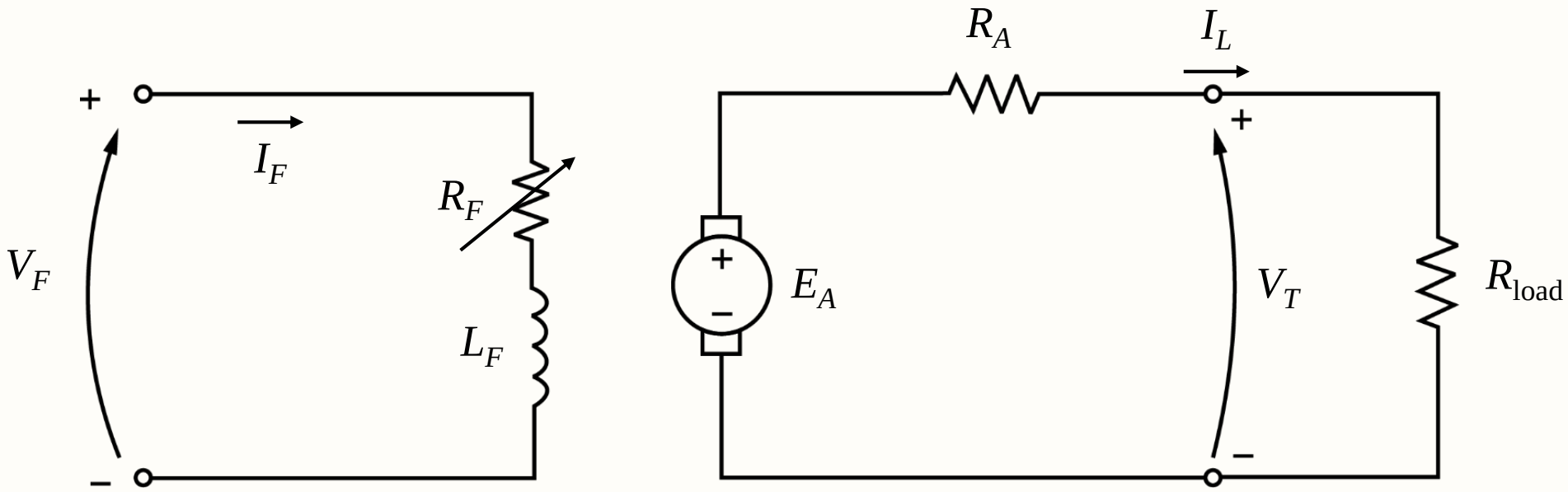
Control of Terminal Voltage of a Separately Excited DC Generator

- The terminal voltage can be controlled by changing the internal generated voltage of the machine, as:

$$V_T = E_A - I_A R_A \Rightarrow \text{If } E_A \uparrow \Rightarrow V_T \uparrow$$

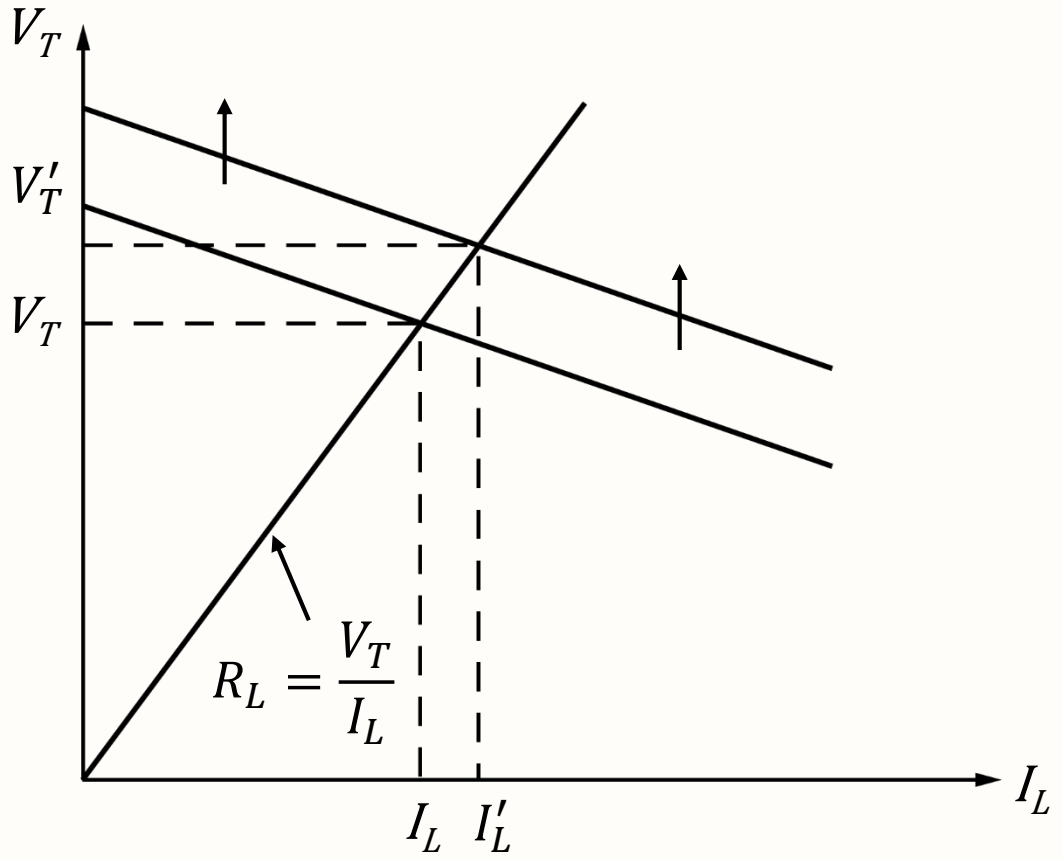
- Since $E_A = K\phi\omega_m$, there are two ways to control E_A:
  - Change ω_m : if $\omega_m \uparrow \Rightarrow E_A \uparrow \Rightarrow V_T \uparrow$
  - Change I_F : if $R_F \downarrow \Rightarrow I_F \uparrow \Rightarrow \phi \uparrow \Rightarrow E_A \uparrow \Rightarrow V_T \uparrow$
- In many applications, the speed range of the prime mover is quite limited, so the terminal voltage is most commonly controlled by changing the field current.

Control of Terminal Voltage of a Separately Excited DC Generator



A separately excited DC generator with a resistive load.

Control of Terminal Voltage of a Separately Excited DC Generator



Effect of a decrease in field resistance on the output voltage of the DC generator.

Non-linear Analysis of a Separately Excited DC Generator

- If a machine has armature reaction AR, its flux will be reduced with each increase in load, causing the internal generated voltage to decrease.
- The output voltage of the generator with AR is determined using the magnetisation curve of the generator.
- The total mmf in the generator is the field circuit mmf less the mmf due to AR.

Nonlinear Analysis of a Separately Excited DC Generator

- $mmf_{net} = N_F I_F^* = N_F I_F - mmf_{AR}$

  Then, $I_F^* = I_F - \frac{mmf_{AR}}{N_F}$

  where I_F^*= equivalent field current that will produce the same output voltage as the combination of all the mmfs in the machine.

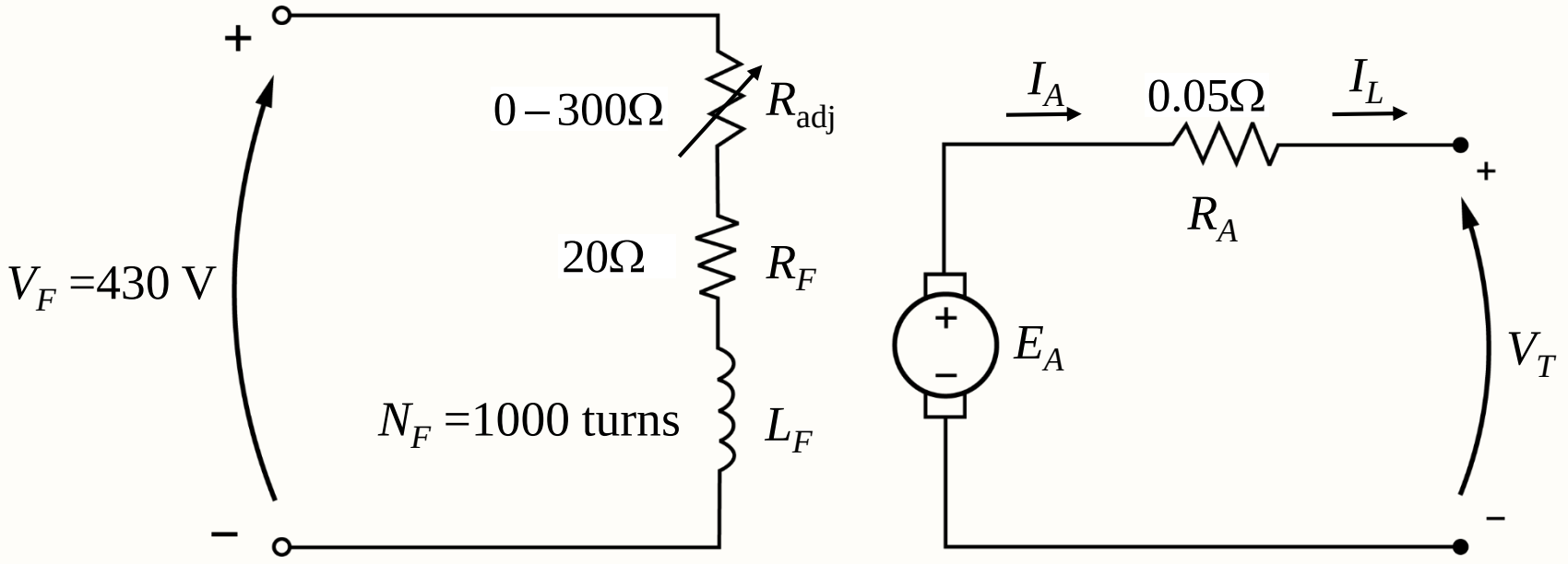
- The resulting voltage E_{A0} can then be determined by locating that equivalent field current on the magnetisation curve.

- The difference between the speed n_0 , at which the magnetisation curve was obtained, and the actual speed n_m of the generator must be taken into account using $\frac{E_A}{E_{A0}} = \frac{n_m}{n_0}$.

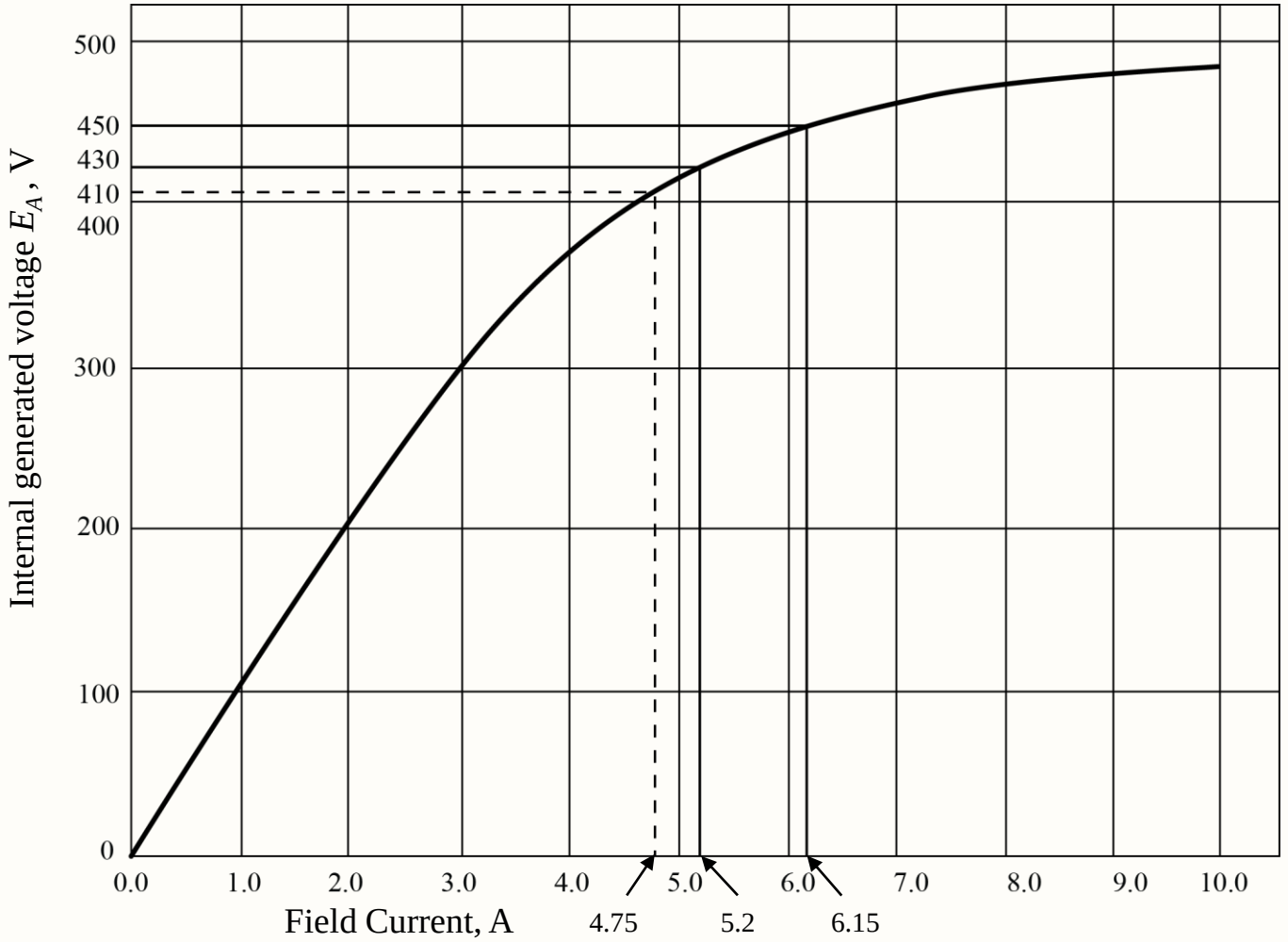
Example 3

- A separately excited DC generator is rated at 172 kW, 430 V, 400 A and 1800 r/min. The magnetisation curve is as shown and the machine characteristics are:

$$R_A = 0.05\Omega,\ V_F = 430\ \text{V},\ R_F = 20\Omega,\ R_{adj} = 0 \text{ to } 300\Omega,\ N_F = 1000 \text{ turns per pole}$$



Example 3



Note: When the field current is zero, E_A is about 3 V.

Example 3

a) If the variable resistor in the generator's field circuit is adjusted to 63 ohms, and the generator's prime mover is driving it at 1600 r/min, what is the generator's no-load terminal voltage?

b) What would its terminal voltage be if a 360 A load were connected to its terminals? Assume that the generator has compensating windings.

c) What would its terminal voltage be if a 360 A load were connected to its terminals, but the generator does not have compensating windings? Assume that its armature reaction at this load is 450 A-turns.

Example 3

d) For the conditions in (b), what adjustment could be made to the generator to restore its terminal voltage to the value found in part (a)?

e) How much field current would be needed to restore the terminal voltage to its no-load value? Assume that the machine has compensating windings. What is the required value of the variable resistor to accomplish this?

(Solutions →)

Example 3 – Solutions

a)

$$I_F = \frac{V_T}{R_F} = \frac{430}{(20+63)} = 5.2 \text{ A}$$

From the magnetisation curve, this much of current would produce an E_{A0} = 430 V at a speed of n_0 = 1800 rpm. Since the generator is rotating at n_m = 1600 rpm, its internal generated voltage will be:

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \Rightarrow E_A = 382 \text{ V}$$

Since $V_T = E_A$ at no-load conditions, the generator's no-load terminal voltage is $V_{T_{NL}}$ = 382 V.

Example 3 – Solutions

b) There is no AR in the machine. If a 360 A load were connected to the generator's terminals, the terminal voltage of the generator would be:

$$V_T = E_A - R_A I_A = 382 - (360)(0.05) = 364 \text{ V}$$

Example 3 – Solutions

c) If a 360 A load were connected to the generator's terminals and the generator has 450 A-turns of AR, then

$$I_F^* = I_F - \frac{mmf_{AR}}{N_F} = 5.2 - \frac{450}{1000} = 4.75 \text{ A}$$

From the magnetisation curve, this much current would produce an E_{A0} = 410 V at 1800 rpm, so the internal generated voltage at 1600 rpm would be:

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \Rightarrow E_A = 364 \text{ V}$$

Therefore, the terminal voltage of the generator would be:

$$V_T = E_A - R_A I_A = 364 - (360)(0.05) = 346 \text{ V}$$

It is lower than before due to the armature reaction.

Example 3 – Solutions

d) Increase E_A by decreasing R_{adj} so as to increase I_F.

e) For $V_T = 382\ \text{V}$, the required E_A is:

$$E_A = V_T + R_A I_A = 382 + (360)(0.05) = 400\ \text{V}$$

To get $E_A = 400\ \text{V}$ at $n_m = 1600\ \text{rpm}$, the internal generated equivalent voltage at $1800\ \text{rpm}$ would be:

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \Rightarrow E_{A0} = 450\ \text{V}$$

From the magnetisation curve, this voltage would require a field current $I_F = 6.15\ \text{A}$, so that

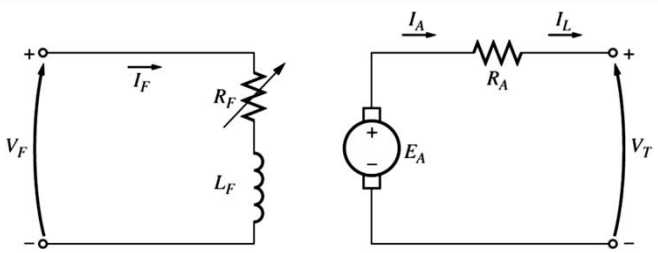
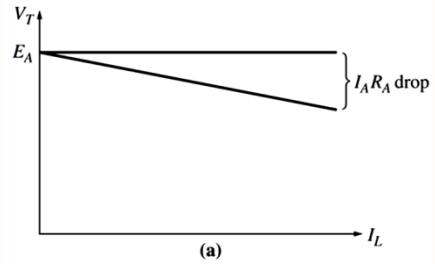
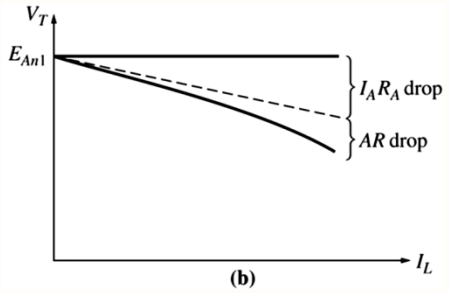
$$20 + R_{adj} = \frac{430}{6.15} \Rightarrow R_{adj} = 50\Omega$$

Summary

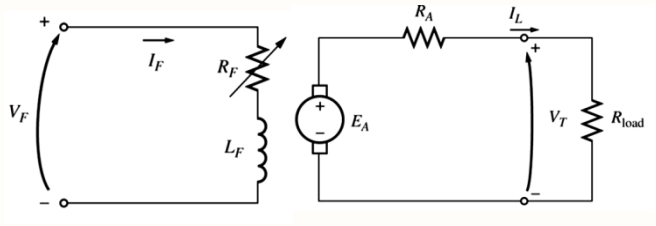
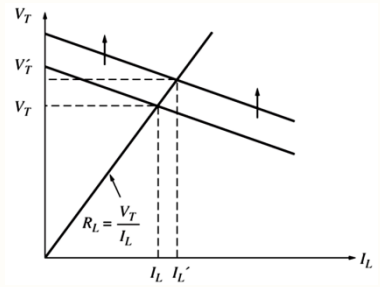
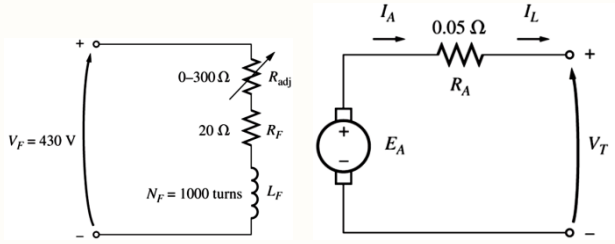
In this lecture, you have learnt:

- The equivalent circuit of a separately excited DC generator.
- The methods used to control the terminal voltage of a separately excited DC generator.
- The analysis of a separately excited DC generator.

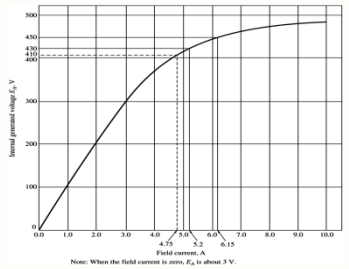
References

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References

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